

# Project Management

Basic term

# Contents

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- Introduction to the course
- Projects, programs, portfolios.
- The value delivery system of an organization.
- Why projects are initiated?
- Who are in a project?
- Concepts of project management.
- Why projects succeed or fail
- Factors affecting the success of a software project (EEF, OPA)
- Organization structure types
- Project management support structure.

Basic term

# What is a project?

- **project(n.)**: "a plan, draft, scheme, design," from Medieval Latin *proiectum* "**something thrown forth**," from *pro-* "**forward**" + combining form of *iacere* (past participle *iactus*) "**to throw**"
  - Build a new house
  - Developing a new product or service

A screenshot of an IELTS Test Report Form. The form contains fields for Candidate Number, Test Date, Candidate Name, and Candidate ID. It also includes a section for Test Results with scores for Listening, Reading, Writing, and Speaking. The form is dated 2008/2009.

The validity of this IELTS Test Report Form can be verified online by entering the candidate's ID and report number.

*A project is a **temporary** endeavor undertaken to create a **unique** product, service or result*

Temporary  
endeavor

Unique (product,  
service, result)





## PROJECT VS PROGRAM VS PORTFOLIO – KEY DIFFERENCES

**What is it?**

### Project

A temporary endeavor undertaken to create a unique project service or result.

### Program

A group of related projects managed in a coordinated way.

### Portfolio

A collection of projects, programs, and other work grouped together.

**What is the goal?**

Delivery of a specific output.

Realizing business-wide benefits.

Meeting strategic business objectives.

**What is the focus?**

Deliverables

Benefits

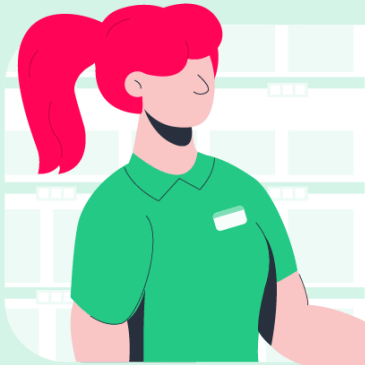
Objectives

**Is it time-restricted or ongoing?**

Time-restricted

Time-restricted

Ongoing



## Retail

Creation of  
a new electric drill.

Creation of a line  
of power tools.

All projects and programs  
regarding the products  
sold by a business.

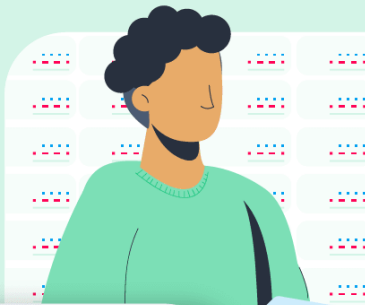


## Construction

Building a house.

All houses built  
in a given area.

All projects and programs  
of the business.



## Software

A new feature  
of the software.

An update containing  
multiple features  
of the software.

All projects and programs  
concerning the product.

# Program & Portfolio

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## Program

- A group of **related projects**, subprograms, and program activities managed in a coordinated way to obtain benefits not available from managing them individually

## Portfolio

- Projects, programs, subportfolios, and operations managed as a group to achieve **strategic objectives**



# Examples about program



# Examples about portfolio

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Luxury housing



Low cost housing



Villas



Towers



Roads



Bridges



Railway routes



Cricket stadium

A consulting company wanted to increase the amount of billable time for each consultant, so it started several company-wide programs to help consultants to get more productivity out of each year.

.....

A university wanted to build admissions websites for all of its departments. It realized that all of the sites would be feeding into the same registration interface and decided to manage all of them together in order to save time.

.....



A company wanted to switch from a paper-based Human Resources group to a software-based one. It spent some time looking into the best software packages for the job, and decided to manage all of the HR functions together since it needed the same people to help with all of the work.

.....

A company wanted to build a better reporting interface so that it could have more accurate data on year-end goals.

.....

A software game company wanted to build up its online presence. It started several marketing and sales initiatives, created some new games, and rewrote some old ones in order to reach more gamers online.

.....

A construction company bid on several parking garage projects at the same time. It won one of the bids, and built the garage a month under schedule and \$5,000 under budget.

.....

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## Portfolio

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## Program

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## Portfolio

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A university wanted to build admissions websites for all of its departments. It realized that all of the sites would be feeding into the same registration interface and decided to manage all of them together in order to save time.

## Program

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A company wanted to build a better reporting interface so that it could have more accurate data on year-end goals.

## Project

---

A construction company bid on several parking garage projects at the same time. It won one of the bids, and built the garage a month under schedule and \$5,000 under budget.

## Project

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# MANAGEMENT

## PROJECT vs OPERATIONS



Temporary Endeavour



Ongoing Routine



Unique Output



Repetitive Output

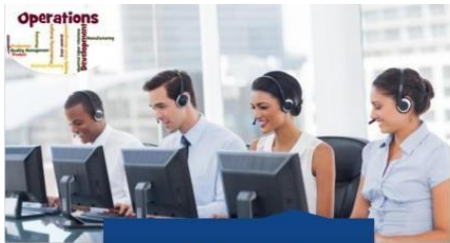


Ends once objectives are achieved



Establishes ongoing objectives

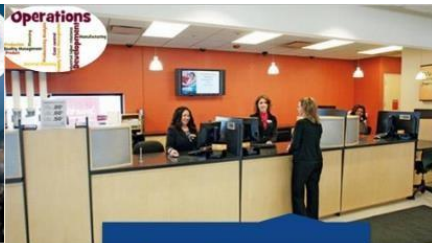
# Project & Operation



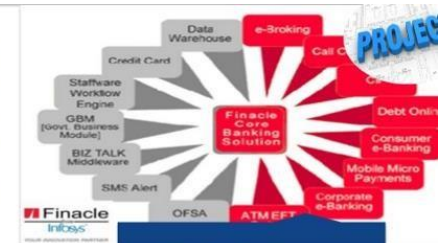
Call Center



Moon Mission



Bank Reception



Bank Software



Bridge Construction



Filling Gas@Petrol Pump



Upgrading to new OS



CAR Manufacturing



## Exercise

1. Building an extension on a house

☐

Operation

☐

Project

2. Shelving books at the library

☐

Operation

☐

Project

3. Baking a wedding cake

☐

Operation

☐

Project

4. Stapling programs for a play

☐

Operation

☐

Project

5. Watering your plants twice a week

☐

Operation

☐

Project

6. Walking the dog every day

☐

Operation

☐

Project

7. Knitting a scarf

☐

Operation

☐

Project

8. Making a birdhouse

☐

Operation

☐

Project

9. Changing your air filters every six months

☐

Operation

☐

Project

10. Running an assembly line in a toy factory

☐

Operation

☐

Project

11. Organizing a large conference

☐

Operation

☐

Project

12. Going to the gym three times a week

☐

Operation

☐

Project

1. Building an extension on a house



Operation



Project

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Operation



Project

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Operation



Project

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Operation



Project

12. Going to the gym three times a week



Operation



Project

# Why organizations initiate a project?

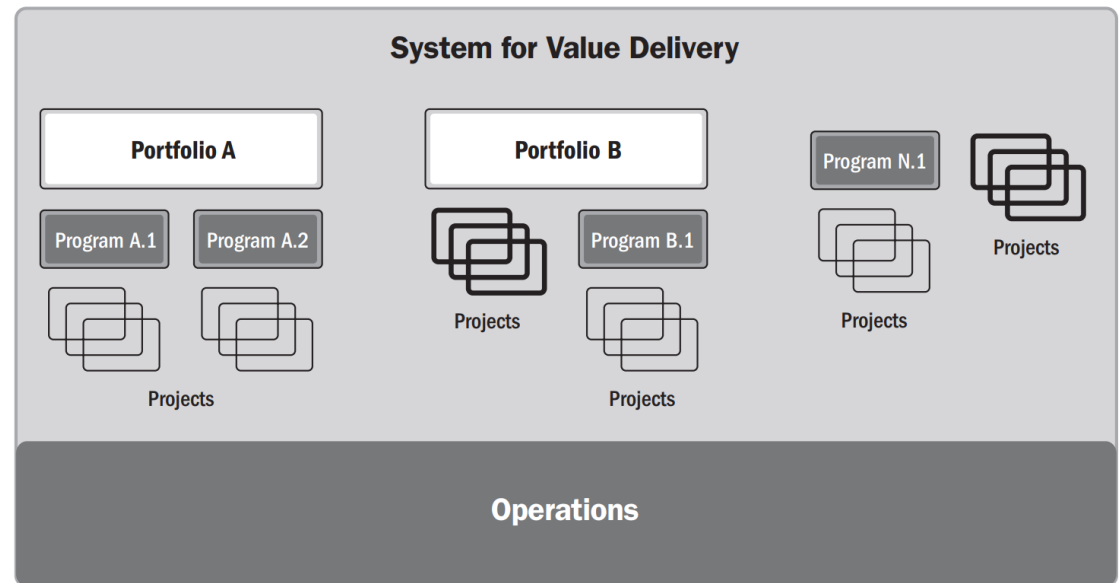
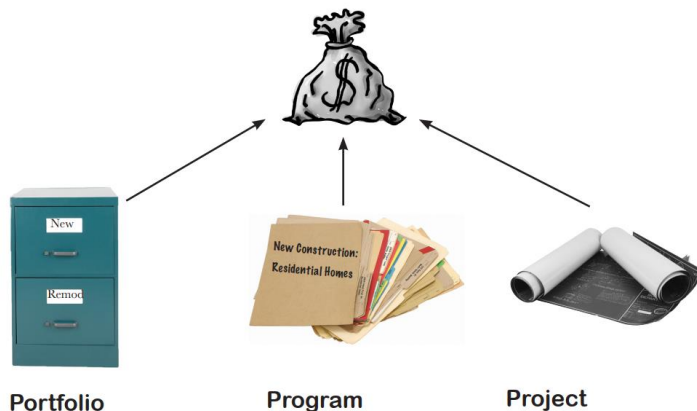
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- Organizations create **value** for **stakeholders**.
- **Value**: the worth, importance, or usefulness of something.
  - **Business value**: determined with financial metrics, such as the benefits less the cost of achieving those benefits.
  - **Societal value**: can include the contribution to groups of people, communities, or the environment.
- Examples of creating value:
  - Creating a new product, service, or result that meets the needs of customers or end users;
  - Creating positive social or environmental contributions;
  - Improving efficiency, productivity, effectiveness, or responsiveness;
  - Enabling the changes needed to facilitate organizational transition to its desired future state;
  - Sustaining benefits enabled by previous programs, projects, or business operations.



# System for Value delivery

- A collection of strategic business activities aimed at building, sustaining, and/or advancing an organization.



# Outcome, Benefits & Value

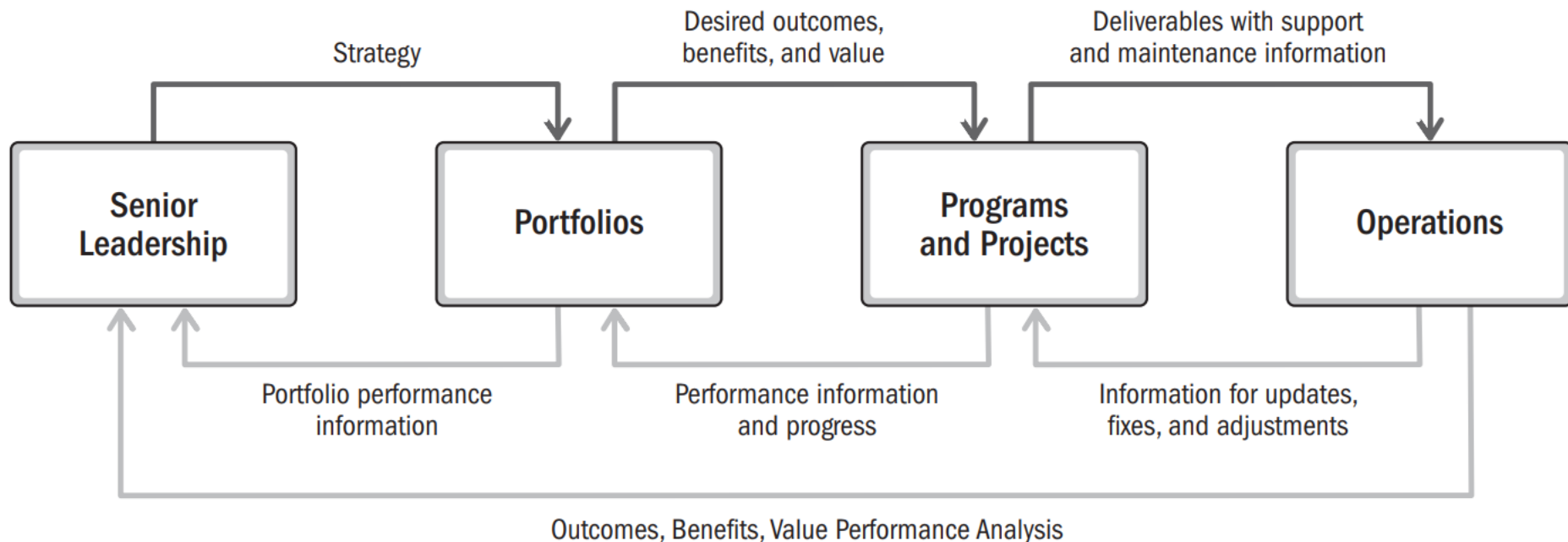
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| Term     | Definition                                                                               | Example                                                                               |
|----------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Outputs  | Specialized product/ deliverable                                                         | New sales system                                                                      |
| Outcomes | The end result or consequence of a process or a project.                                 | Sales order are processed more quickly with few error                                 |
| Benefits | The outcomes create benefits, which are gains realized by the organization.              | Costs are reduced by 10%, sale order are +15%, revenue +10%                           |
| Values   | Benefits, in turn, create value, which is something of worth, importance, or usefulness. | Smooth customer experience.<br><b>“Value is in the eye of the Buyer, not Seller”.</b> |

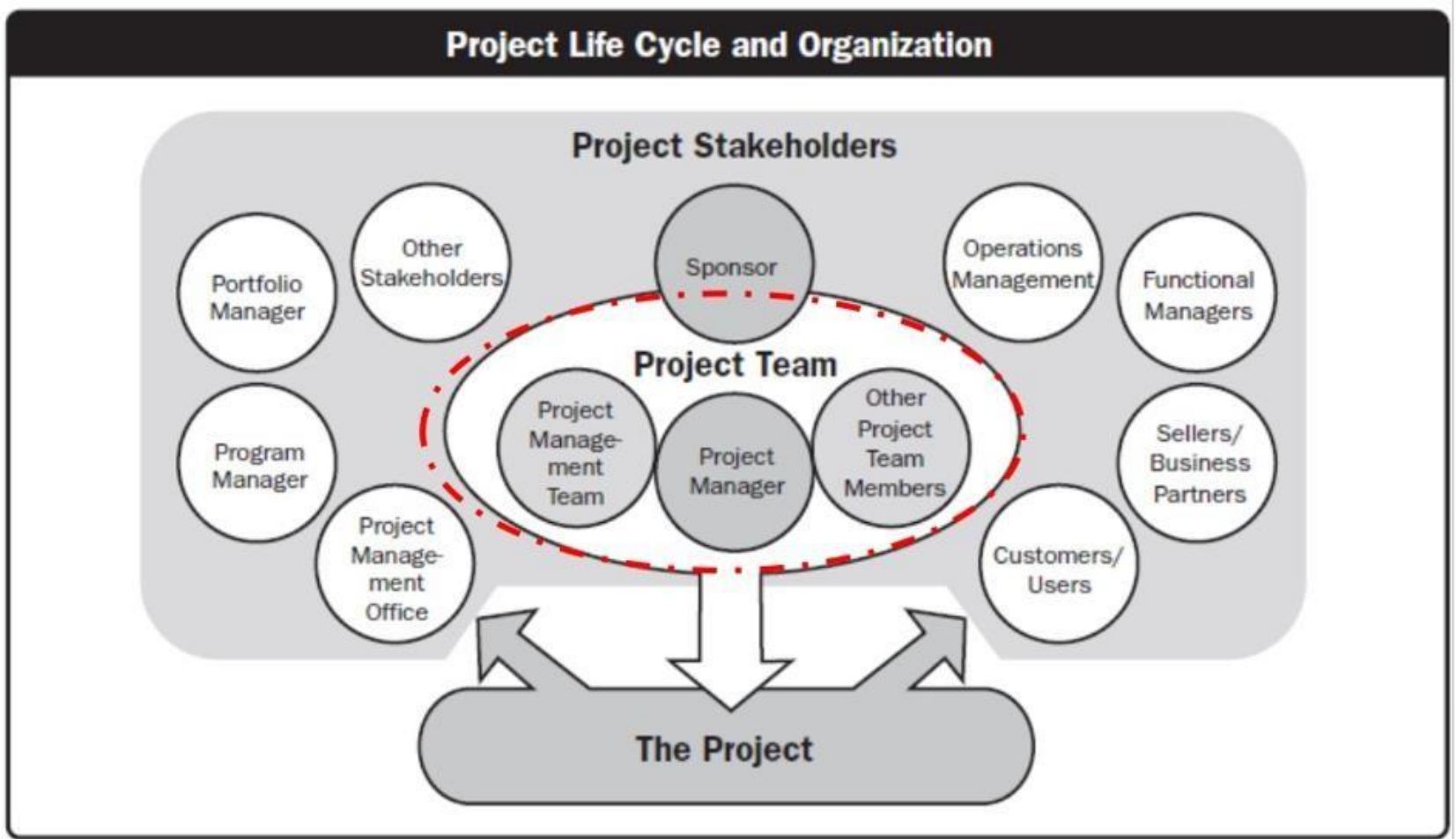
# Information flow

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- **Information and feedback** are shared consistently among all components, keeping the system **aligned with strategy** and **attuned to the environment**.



# Stakeholders



# Stakeholders

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- A **stakeholder** is someone whose interest may be **positively** or **negatively impacted** by the project.
- Key stakeholders:
  - The project manager
  - Customer
  - Performing organization
  - Project Team
  - Project Management Team
  - Sponsor
  - Project Management Office
  - Other influencers



# Project Team & Project Manager

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## Project team

- A set of individuals performing the work of the project to achieve its objectives.

## Project Manager

- The person assigned by the performing organization to lead the team that is responsible for achieving the project objective.
- Applies project management knowledge and uses personal and leadership skills to achieve project success.



# Summary

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- What is a project?
- Why does organization create a project?
- Who is related to a project?

# Summary

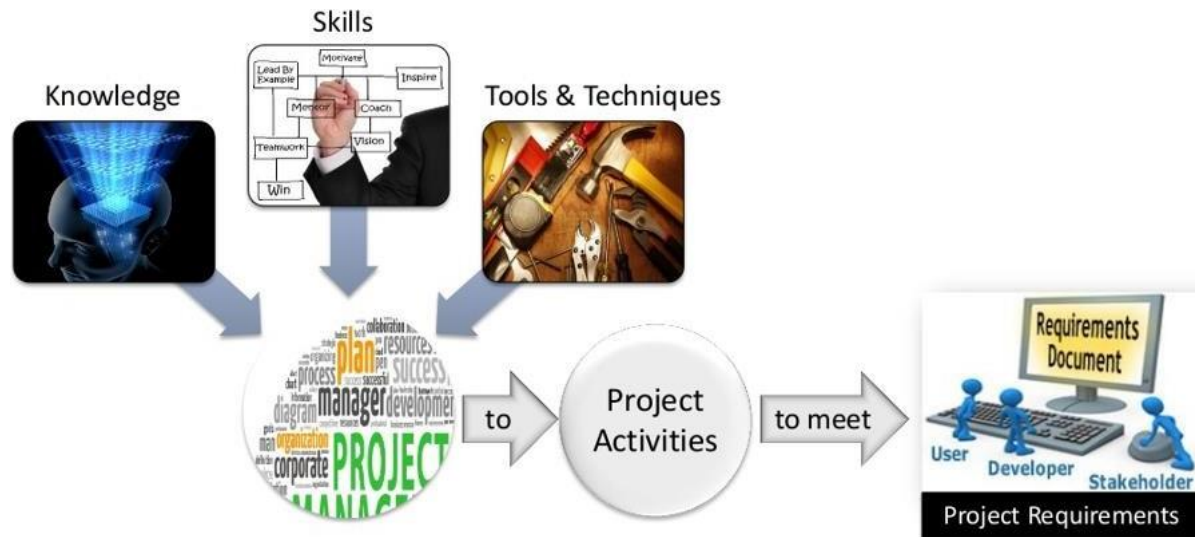
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- What is a project?  
⇒ *temporary endeavor undertaken to create a unique product, service or result*
- Why does organization create a project?  
⇒ *Value*
- Who is related to a project?  
⇒ *Stakeholders*

# Project Management

# Project Management

- The application of knowledge, skills, tools & techniques to project activities to meet project requirements.



# Project Management

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## Project

- A project is a temporary endeavor undertaken to create unique product, service or result

## Project Management

- The application of knowledge, skills, tool & techniques to project activities to meet project requirements

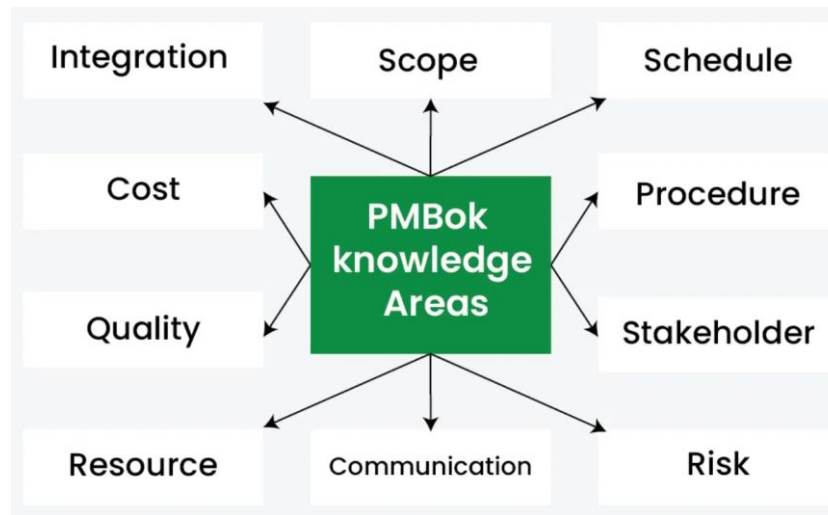
## Project Manager

- The Person assigned by the Performing Organization to lead the team to achieve the project Objectives

# What is PMBOK

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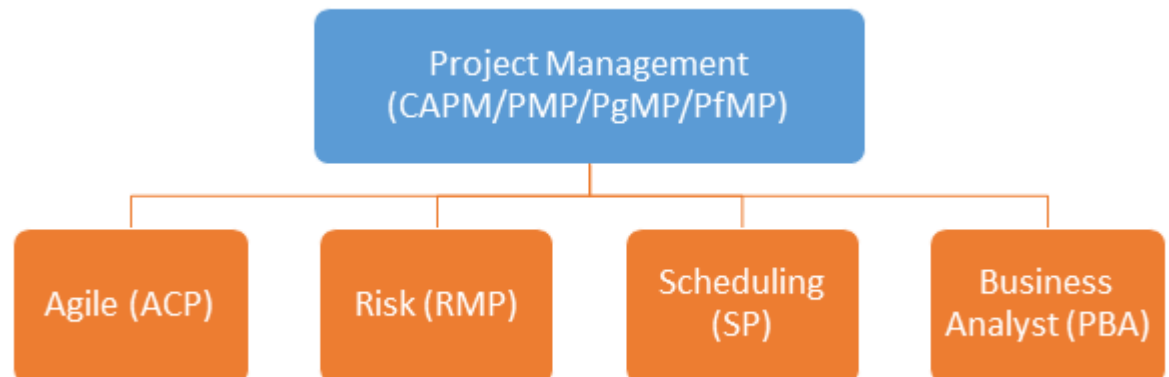
- PMBOK (Project Management Body of Knowledge): entire collection of **processes**, **best practices**, **terminologies**, and **guidelines** that are accepted as standard within the project management industry published by The Project Management Institute (PMI)



# Project Management Institute (PMI)

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- Global organization that provides **standards**, **education**, and **certifications** in project management

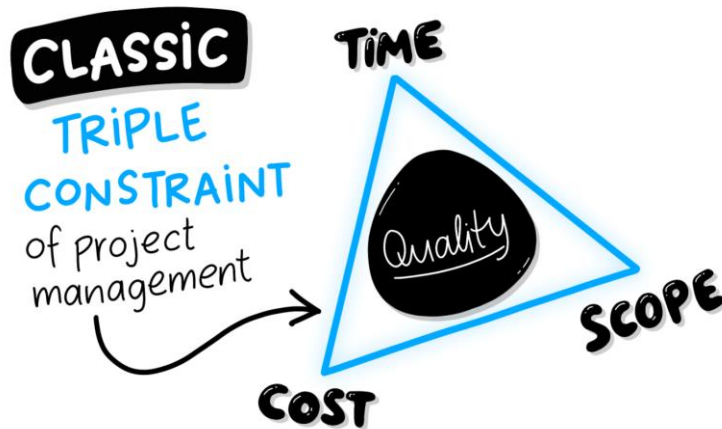


**Successful Project**

# What is a successful project

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- Measured in terms of delivering the **intended outcomes** within **constraints**.



Great software development delivers...

What the customer needs, otherwise called the software requirements. We'll talk more about requirements in the next chapter...

## What is needed,

When we agreed with the customer that the software would be finished.

{ **On Time,**

and

## On Budget

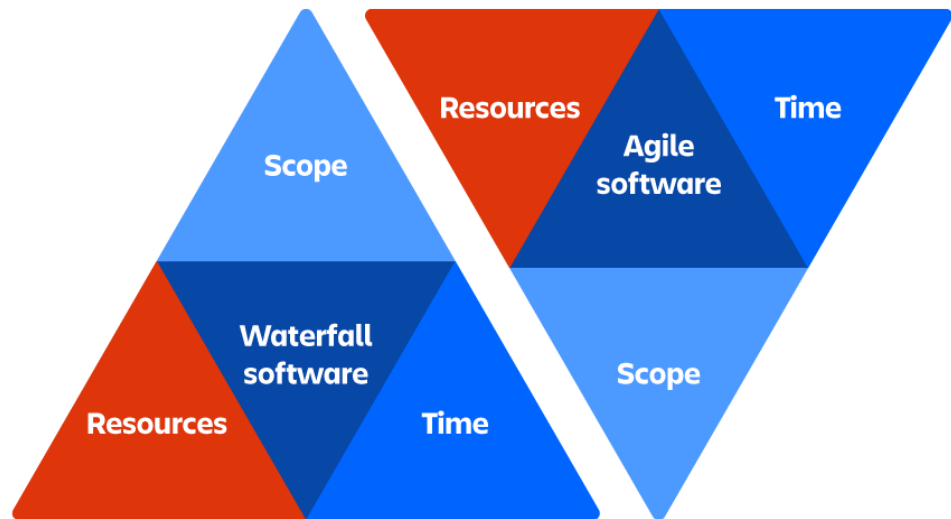
Not billing the customer for more money than was agreed upon.

# Evolution of the triangle

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**Fixed**

**Estimated**

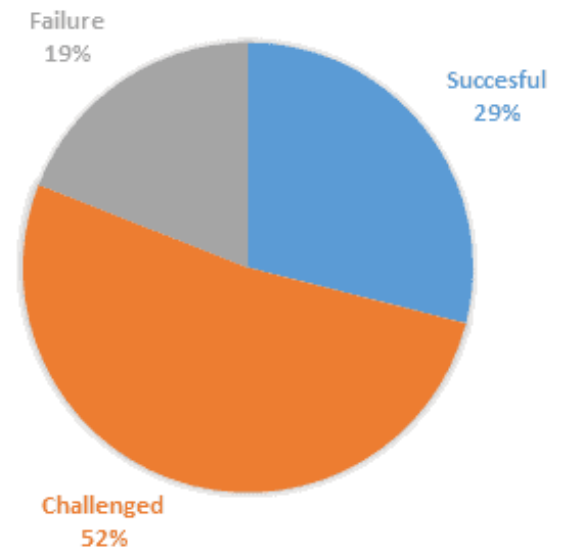


# Why do we need to manage project?

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SUCCESS RATE



Your project will need to get done on schedule.

Your project will always have to stay within a budget.

You need to manage the scope of work you do for the project.

You have to have the people and materials to get the work done.

If your product doesn't do what it's supposed to do, you won't succeed.

Unexpected obstacles can wreck your project if you don't deal with this one.





The project was running late, so the project manager decided to release it on time even though it was missing some of its features.

**Constraint affected:**

.....

The team wanted to add more testers to find defects, but the project manager overruled them.

**Constraint affected:**

.....

A construction project manager assumed that the weather would cooperate with the plans to complete the job, but thunderstorms have derailed the project.

**Constraint affected:**

.....

The company didn't have enough money to invest in the project, so they had to draft people from other departments to work part-time to get the job done.

**Constraint affected:**

.....

About halfway through the project, the PM realized that the money was running out faster than expected. She went through the schedule to try to find ways to move up the deadline.

**Constraint affected:**

.....

The project manager didn't take software license fees into account, which caused the budget to balloon out of control.

**Constraint affected:**

.....

The project was running late, so the project manager decided to release it on time even though it was missing some of its features.

**Constraint affected:**

**scope**

The PM stuck to the original budget and schedule, but released a product that wasn't complete. That means the scope was affected.

The team wanted to add more testers to find defects, but the project manager overruled them.

**Constraint affected:**

**quality**

Any time you're talking about tests and defects, you're talking about quality.

A construction project manager assumed that the weather would cooperate with the plans to complete the job, but thunderstorms have derailed the project.

**Constraint affected:**

**risk**

Whenever you make assumptions about a project, you're introducing risk.

It's the project manager's job to always look after the bottom line.

The company didn't have enough money to invest in the project, so they had to draft people from other departments to work part-time to get the job done.

**Constraint affected:**

**resources**

Resources are people or materials that you need for your project, and when you cut corners you end up straining them.

About halfway through the project, the PM realized that the money was running out faster than expected. She went through the schedule to try to find ways to move up the deadline.

**Constraint affected:**

**time**

There are lots of ways to change how long it'll take to do your project, but sometimes there simply isn't enough time.

The project manager didn't take software license fees into account, which caused the budget to balloon out of control.

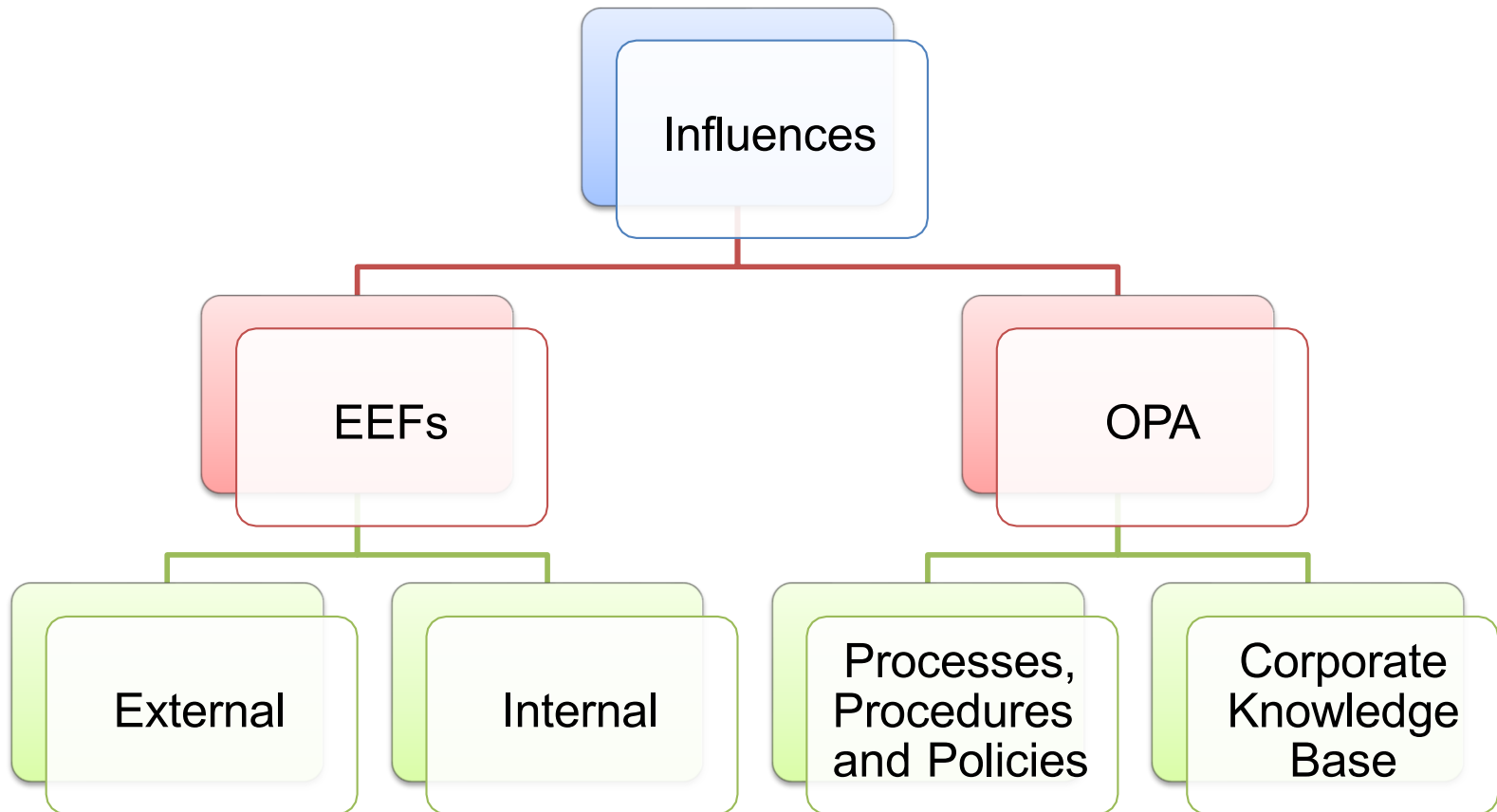
**Constraint affected:**

**cost**

# Project Management Influences

# Project Influential Factors

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# Organizational process assets

## Process

The way your organization does its work.

## Policies

The rules your organization follows when it works on a project.



## Procedures

The practical guides that describe the steps each team will follow when they do work. These include:

- Templates
- Vendor/supplier lists
- Change control guidance
- Financial control guidance
- Estimation methods

***Knowledge repositories:*** all of the lessons learned from past teams

# Organizational Process Assets

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- Include any process assets (artifact, practice, or knowledge...) from any or all of the performing organizations involved in the project that can be used to execute or govern the project.
- OPAs are internal to the organization, the project team members may be able to update and add to the organizational process assets as necessary throughout the project.

| Processes, Procedures and Policies                                                                 | Corporate Knowledge Base                                                                                                                                              |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Policies</li><li>• Processes</li><li>• Procedure</li></ul> | <ul style="list-style-type: none"><li>• Financial databases</li><li>• Historical information</li><li>• Lessons learned</li><li>• Issue and defect databases</li></ul> |

Not updated as part of the project work.

Updated throughout the project

# Enterprise environmental factors

## People

The skills and organizational culture where you work.

## Market

The way your company is performing in the market can affect the way you manage your project.



## Risk tolerance

Some companies are highly tolerant of risk and some are really risk-averse.

## Standards

Some companies depend on government standards to run their business and when they change, it can have a big impact.

## Databases

Where your company stores its data can make a big difference in the decisions you make on your project.

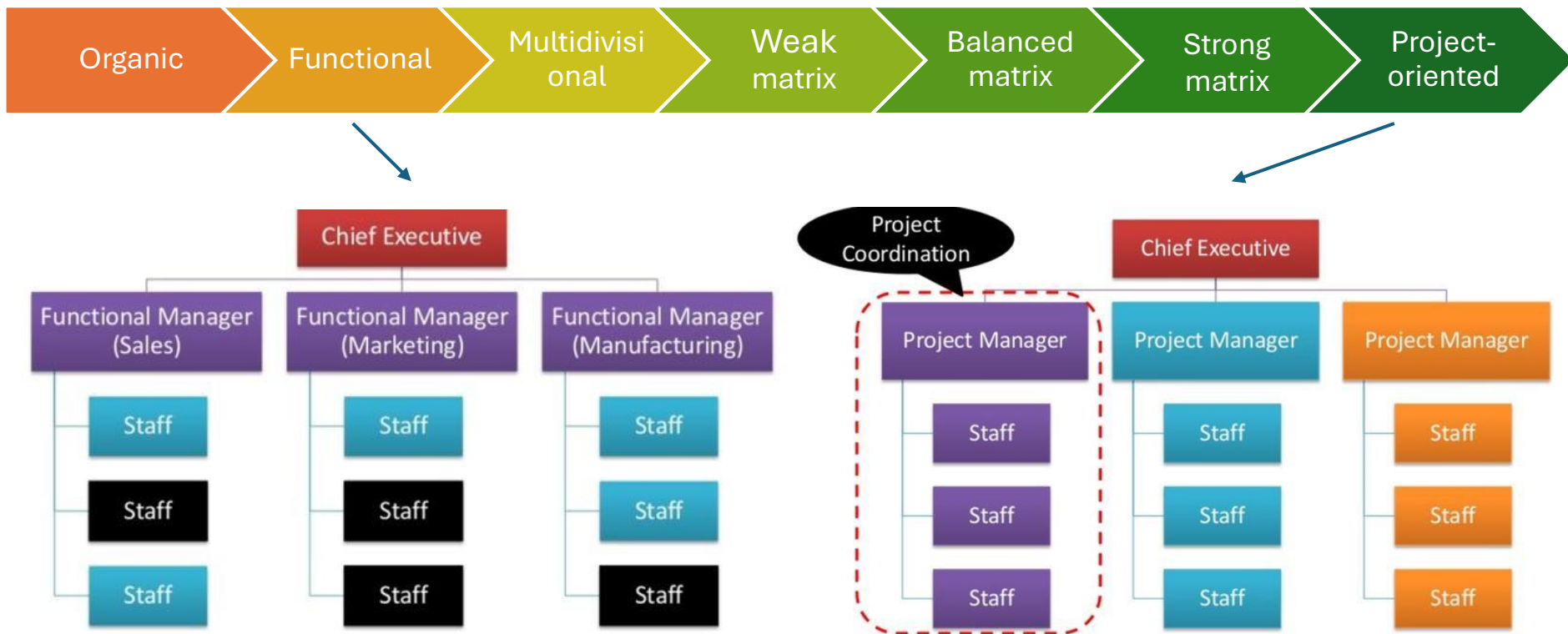
# Enterprise Environmental Factors

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| Internal                                                                                                                                                                                                                                                                                                                    | External                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Organizational culture, structure and governance</li><li>• Geographic distribution of facilities and resources;</li><li>• Infrastructure</li><li>• Project management information system</li><li>• Existing human resources</li><li>• Company work authorization systems;</li></ul> | <ul style="list-style-type: none"><li>• Marketplace conditions;</li><li>• Social and cultural</li><li>• Legal</li><li>• Commercial database</li><li>• Academic research</li><li>• Government or industry standards</li><li>• Political climate;</li></ul> |

# Types of organizational structures

*The project manager has the most authority and power in a project-oriented organization.*



| STRUCTURE TYPE | ORGANIC / SIMPLE                                                                                 | FUNCTIONAL                                                                                          | WEAK MATRIX                                                                                    | BALANCED MATRIX                                                                             | STRONG MATRIX                                                                 | PROJECTIZED                                                                                               |
|----------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| PM ROLE        | Rarely exists or very informal                                                                   | Little to no authority; acts as a coordinator if needed                                             | Coordinator with limited authority; needs FM approval to access resources                      | Shares authority with FM; plans and manages project scope and schedule                      | Strong authority over project budget, team, and decisions                     | Full authority; leads the team and project end-to-end                                                     |
| FM ROLE        | Full control over tasks and people                                                               | Full control over resources, schedules, and decision-making                                         | Maintains main control, but collaborates with PM on tasks                                      | Provides technical expertise and manages team members functionally                          | Acts as technical advisor or resource provider                                | Rarely involved                                                                                           |
| SUITABLE FOR   | Small team, informal communication, flexible operations                                          | Emphasizes specialization, clear hierarchy, easy to manage expertise                                | Early stage of shifting toward project-based work; limited PM capacity                         | Need for cross-department collaboration without breaking department structure               | Projects are complex, need fast decisions, specialized project teams          | Business is fully project-driven, often unique/customized work                                            |
| EXAMPLE        | A 30-person garment workshop where the production manager assigns tasks directly based on orders | Auto parts company with separate departments: design, manufacturing, QA – each handles its own work | Electronics company where PM needs approval from each FM to get engineers for a product launch | Food packaging company launching a new product with coordination between PM and departments | Home appliances company with a strong R&D team managed by project-focused PMs | Medical equipment company delivering custom solutions per hospital contract, with a dedicated PM and team |

## ORGANIC

- Project management is handled as a parttime responsibility for someone on a project team.
- Project managers are usually contributors to the project who take on the added responsibility of helping stakeholders understand project status.
- In an organic organization, it's likely that the project manager has little responsibility for budget decisions.
- Usually those kinds of decisions are made by the company owner.

## FUNCTIONAL

- Project management decisions need to be cleared with functional managers.
- Project managers are the functional managers in getting the work done.
- Project managers spend a lot of time doing administrative tasks and often work as PMs only part of the time.
- You're likely to find **project expeditors** in functional organizations.

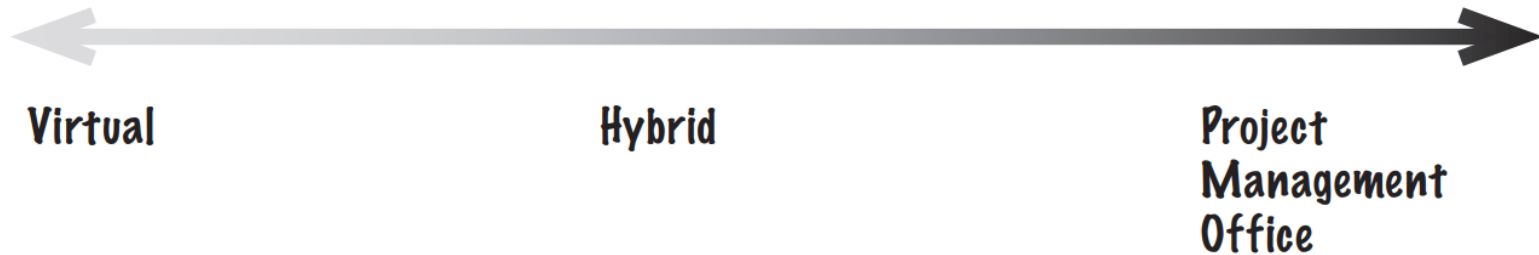
## MULTIDIVISIONAL

- PMs might operate in a completely different way from division to division.
- PMs are often **project coordinators** or parttime **expeditors** who actually work on the project team as a full-time contributor but take on the added responsibility of communicating project status.
- Decisions about budget or resource allocation need to be made by functional managers.

| WEAK METRIX                                                                                                                                                                                                                                                                                                                                              | BALANCED MATRIX                                                                                                                                                                                                                                                  | STRONG MATRIX                                                                                                                                                                                                                                                                                                                                    | PROJECTIZED                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>PMs have some authority, but they aren't in charge of the resources on a project.</li> <li>Major decisions still need to be made with the functional manager's cooperation or approval.</li> <li><b>Project expeditors</b> and <b>project coordinators</b> can work in weak matrix organizations, too.</li> </ul> | <ul style="list-style-type: none"> <li>Project managers share authority with the functional managers.</li> <li>PMs run their people-management decisions by the functional manager, but the functional manager runs project decisions by the PM, too.</li> </ul> | <ul style="list-style-type: none"> <li>Project managers have more authority than functional managers, but the team still reports to both managers.</li> <li>The team might be judged based on performance on their projects, as well as on their functional expertise. In a strong matrix, delivery of the project is most important.</li> </ul> | <ul style="list-style-type: none"> <li>Teams are organized around projects. When a project is done, the team is released, and the team members move on to another project.</li> <li>The project manager makes all of the decisions about a project's budget, schedule, quality, and resources.</li> <li>The PM is responsible for the success or failure of the project.</li> </ul> |

# Supporting structures

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- PMO is a coordinating entity for project management:
  - In a Functional organization: it helps coordinate across departments.
  - In a Matrix organization: it supports resource management and methodologies.
  - In a Projectized organization: it ensures standardization and reporting.
- Virtual structure enables **cross-boundary collaboration**

# Project Management Office (PMO)

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- PMO is a department in the organization.
- A **management structure** that standardizes project-related governance processes & facilitates sharing of resources, methodologies, tool & techniques





## The difference between a PMO and a project manager



**PM**



**PMO**

**One person**

**1**

**Team of people**

**Individual projects**

**2**

**Multiple projects and portfolio**

**Execution and delivery**

**3**

**Strategy and oversight**

# Project Management Office (PMO)

- A primary function of a PMO is to support project managers in a variety of ways, there are several types of PMOs:

| Type of PMO              | Supportive                                                                                                     | Controlling                                                                                   | Directive                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Degree of Control        | Low                                                                                                            | Moderate                                                                                      | High                                                                                       |
| Roles and responsibility | Supplying templates, best practices, training, access to information, and lessons learned from other projects. | Compliance to specific templates, forms, and tools; and conformance to governance frameworks. | Directly managing the projects.<br>Project managers are assigned by and report to the PMO. |



# Software company structure

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- Matrix Structure (most common in medium to large software companies)
  - Encourages collaboration across departments.
  - Efficient use of skilled resources.
  - Suits Agile and cross-functional teams.
- Projectized Structure (popular in startups and product-driven teams)
  - Prioritizes product speed and flexibility.
  - Faster decision-making. Clear ownership and focus.
  - Highly adaptable to market changes.
- Functional Structure (used in traditional or early-stage companies)
  - Simple and easy to manage in small teams.
  - Slower communication across departments; not ideal for rapid software iteration.

# Software company structure

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- Modern Trend: hybrid of **Matrix + Agile/PMO + Virtual**
  - Agile teams (Scrum or Kanban).
  - A PMO to align strategy and process.
  - A virtual structure to support remote and distributed workforces.

| Structure        | Key Features                                                                                                                          | Advantages                                                                                                                 | Limitations                                                                                                                | Best for                                      |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Functional       | <ul style="list-style-type: none"> <li>- Organized by departments (Dev, QA, Design, etc.)</li> <li>- Clear hierarchy</li> </ul>       | <ul style="list-style-type: none"> <li>✓ Clear roles and career paths</li> <li>✓ Efficient for repetitive tasks</li> </ul> | <ul style="list-style-type: none"> <li>✗ Siloed communication</li> <li>✗ Slow project response time</li> </ul>             | Small or early-stage companies                |
| Matrix           | <ul style="list-style-type: none"> <li>- Dual authority: functional + project managers</li> <li>- Resource sharing</li> </ul>         | <ul style="list-style-type: none"> <li>✓ Good resource utilization</li> <li>✓ Encourages collaboration</li> </ul>          | <ul style="list-style-type: none"> <li>✗ Conflict in reporting lines</li> <li>✗ Slower decisions</li> </ul>                | Medium to large companies using Agile         |
| Projectized      | <ul style="list-style-type: none"> <li>- Teams organized around products/projects</li> <li>- Project manager has authority</li> </ul> | <ul style="list-style-type: none"> <li>✓ Fast execution</li> <li>✓ Clear accountability</li> </ul>                         | <ul style="list-style-type: none"> <li>✗ Resource duplication</li> <li>✗ Less specialization across projects</li> </ul>    | Startups, product-driven environments         |
| Virtual          | <ul style="list-style-type: none"> <li>- Remote or globally distributed teams</li> <li>- Tech-enabled coordination</li> </ul>         | <ul style="list-style-type: none"> <li>✓ Cost-effective</li> <li>✓ Access to global talent</li> </ul>                      | <ul style="list-style-type: none"> <li>✗ Requires strong communication and tools</li> <li>✗ Harder team bonding</li> </ul> | Remote-first, global software companies       |
| PMO (supporting) | <ul style="list-style-type: none"> <li>- Central body governing project methods</li> <li>- Supports any structure</li> </ul>          | <ul style="list-style-type: none"> <li>✓ Standardizes processes</li> <li>✓ Improves project oversight</li> </ul>           | <ul style="list-style-type: none"> <li>✗ Can become bureaucratic if too rigid</li> </ul>                                   | Companies with multiple simultaneous projects |

# Summary

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- Which factors affect a project?  
⇒ *EEFs & OPAs*
- Typical organization structures? What is main difference?  
⇒ *Functional -> Matrix -> Projectized*  
⇒ *Authority chain*
- What is PMO?  
⇒ *Structure that standardizes project-related governance processes & facilities sharing of resources, methodologies, tool & techniques*